Yadong Liu

Curriculum Vitæ

School of Mathematical Sciences
Nanjing Normal University
No.1 Wenyuan Road, Qixia District
Nanjing 210023, P. R. China
ydliu@njnu.edu.cn
Last Updated: October 17, 2026

"Ein Mathematiker ist eine Maschine, die Kaffee in Theoreme umwandelt" - Alfréd Rényi

Personal Information

Gender	Male
Birth	November 27, 1996 (*Yancheng*, Jiangsu, P.R. China)
Nationality	Chinese
Homepage	https://yadongliu-math.github.io/

Appointment

2024.01 – Now	**Lecturer**, **Nanjing Normal University**, Nanjing, China
2023.08 – 2023.12	**Wissenschaftlicher Mitarbeiter (Postdoc)**
	University of Regensburg, Regensburg, Germany
	Supported by startup funding of RTG 2339 *Interfaces, Complex Structures, and Singular Limits* of the German Research Foundation (DFG)
2020.10 – 2023.07	**Wissenschaftlicher Mitarbeiter (Research Assistant)**
	University of Regensburg, Regensburg, Germany
	In RTG 2339 *Interfaces, Complex Structures, and Singular Limits* of the German Research Foundation (DFG)

Education

2020.10 – 2023.07	**Dr. rer. nat. in Mathematics**
	University of Regensburg, Regensburg, Germany
	Dissertation: Analysis of fluid-structure interaction problems with growth
	Grade: *magna cum laude*
	Supervisors: Prof. Dr. Helmut Abels & PD Dr. Maria Neuss-Radu
	Defense Committee: Bernd Ammann, Helmut Abels, Matthias Hieber, Harald Garcke, Bogdan Matioc.
2018.09 – 2020.06	**M.Sc. in Mathematics**
	Nanjing University of Information Science and Technology, Nanjing, China
2015.09 – 2018.06	**B.Sc. in Mathematics and Applied Mathematics**
	Nanjing University of Information Science and Technology, Nanjing, China
2014.09 – 2015.07	**Undergraduate study in Physics**
	Nanjing University of Information Science and Technology, Nanjing, China

Research Interests

Analysis of nonlinear partial differential equations

- Fluid-structure interactions: bulk-surface, bulk-bulk, well-posedness, incompressible/compressible, singular limits (low Mach number, high Reynolds number)
- Two-phase flows: diffuse interface models, viscoelastic fluids, well-posdeness, long time dynamics, sharp interface limit

Publications & Preprints

Preprints

16. **Y. Liu**, Global strong solutions without contact for a viscous fluid interacting with a purely elastic beam, (2026) arXiv: *2610.22090*.
15. H. Abels, M. Fei, **Y. Liu** and M. Moser, Higher Order Convergence for the Sharp Interface Limit of 3D Navier–Stokes/Allen–Cahn Systems, (2026) arXiv: *2606.30532*.
14. M. Fei, X. Fei, **Y. Liu** and H. Wu, Global weak solutions to a diffuse-interface model for quasi-incompressible two-phase flows with unmatched densities and singular potential, (2026) arXiv: *2604.26660*.

Publications in Peer-Reviewed Journals

13. M. Fei, X. Fei, D. Han and **Y. Liu**, Weak solutions and incompressible limit of a quasi-incompressible Navier–Stokes/Cahn–Hilliard model for viscous two-phase flows, to appear in ***SIAM J. Math. Anal.*** (2025) arXiv: *2508.08090*.
12. P. Knopf and **Y. Liu**, A Thermodynamically Consistent Free Boundary Model for Two-Phase Flows in an Evolving Domain with Bulk-Surface Interaction, to appear in ***Interface Free Bound.*** (2026) arXiv: *2507.01618*.
11. **Y. Liu**, S. Mitra and Š. Nečasová, Weak solutions and singular limits for a compressible fluid-structure interaction problem with slip boundary conditions, ***SIAM J. Math. Anal.*** **58** (2026), no. 3, 2536–2577. DOI: 10.1137/25M1736827
10. M. Fei, X. Fei, D. Han and **Y. Liu**, Local-in-time existence of strong solutions to a quasi-incompressible Cahn–Hilliard–Navier–Stokes system, ***Zeitschrift für Angewandte Mathematik und Mechanik*** **106** (2026), no. 5, e70443, 1–24. DOI: 0.1002/zamm.70443
9. **Y. Liu** and D. Trautwein, On a diffuse interface model for incompressible viscoelastic two-phase flows, ***J. Nonlinear. Sci.*** **35** (2025), no. 12, 1–71. DOI: 10.1007/s00332-024-10103-w
8. H. Abels, **Y. Liu** and Š. Nečasová, Low mach number limit of a diffuse interface model for two-phase flows of compressible viscous fluids, ***GAMM-Mitteilungen*** (2024), e202470008, 1–15. DOI: 10.1002/gamm.202470008.
7. H. Abels and **Y. Liu**, Short time existence of a quasi-stationary fluid-structure interaction problem for plaque growth, ***Adv. Nonlinear Anal.*** **12** (2023), no. 1, 20230101. DOI: 10.1515/anona-2023-0101.
6. H. Abels and **Y. Liu**, On a fluid-structure interaction problem for plaque growth: cylindrical domain, ***J. Differential Equations*** **345** (2023), 334–400. DOI: 10.1016/j.jde.2022.11.046
5. H. Abels and **Y. Liu**, On a fluid-structure interaction problem for plaque growth, ***Nonlinearity*** **36** (2023), 537–583. DOI: 10.1088/1361-6544/aca5e1
4. Y. Zheng, W. Liu and **Y. Liu**, Equivalence between internal observability and exponential stabilization for suspension bridge problem, ***Ric. Mat.*** **71** (2022), no. 2, 711–721. DOI: 10.1007/s11587-021-00566-4
3. W. Liu, Y. Luan, **Y. Liu** and G. Li, Well-posedness and asymptotic stability to a laminated beam in thermoelasticity of type III, ***Math. Methods Appl. Sci.*** **43** (2020), no. 6, 3148–3166. DOI: 10.1002/mma.6108
2. **Y. Liu** and W. Liu, Blood flow analysis in tapered stenosed arteries with the influence of heat and mass transfer, ***J. Appl. Math. Comput.*** **63** (2020), no. 1-2, 523–541. DOI: 10.1007/s12190-020-01328-5

1. **Y. Liu** and W. Liu, Chaotic behavior analysis and control of a toxin producing phytoplankton and zooplankton system based on linear feedback, ***Filomat*** **32** (2018), no. 11, 3779–3789. DOI: 10.2298/fil1811779l

Academic Activities

Research stays

July–Aug. 2026	Tianyuan Mathematics Research Center, Kunming, China Visiting Scholars Program: *Multiscale dynamics and singular limits of two-phase flows*
Apr.,May 2026	School of Mathematics and Statistics, Anhui Normal University, Wuhu, P.R. China Host Professor: Prof. *Mingwen Fei*, Ph.D.
June 2025	Fakultät Mathematik, Universität Regensburg, Regensburg, Germany Host Professor: Prof. Dr. *Helmut Abels*
Mar. 2025	School of Mathematical Sciences, Fudan University, Shanghai, P.R. China Host Professor: Prof. *Hao Wu*, Ph.D.
Jan.,Nov., 2025	School of Mathematics and Statistics, Anhui Normal University, Wuhu, P.R. China Host Professor: Prof. *Mingwen Fei*, Ph.D.
Oct. 2024	School of Mathematics and Statistics, Anhui Normal University, Wuhu, P.R. China Host Professor: Prof. *Mingwen Fei*, Ph.D.
June 2024	Fakultät Mathematik, Universität Regensburg, Regensburg, Germany Host Professor: Prof. Dr. *Helmut Abels*
Nov. 2023	Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic Host Professor: RNDr. *Šárka Nečasová*, DSc
Oct. 2023	Institute for Analysis, Karlsruhe Institute of Technology, Karlsruhe, Germany Host Professor: JProf. Dr. *Xian Liao*
Aug. 2023	School of Mathematical Sciences, Fudan University, China Host Professor: Prof. *Hao Wu*, Ph.D.
Nov. 2022	Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic Host Professor: RNDr. *Šárka Nečasová*, DSc
Apr.–June 2022	Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic Host Professor: RNDr. *Šárka Nečasová*, DSc

Organisation of meetings

Sept. 2026	The 17th International Conference on *Free Boundary Problems*, Berlin, Germany Co-organized the minisymposium: Fluid-Structure Interactions
Oct. 2022	Annual Meeting of RTG 2339 *IntComSin*, Weltenburg Co-organized and contributed talk
Apr. 2022	Workshop on *Analysis of Fluid and Elastic Bodies Interactions*, Regensburg Co-organized and contributed talk

Invited talks

Mar. 2027	Conference on *Regularity, effective behavior and asymptotic flows for models of fluid mechanics*, Marseille Luminy, France
Feb. 2027	International Conference on the Mathematics of *Interfaces, Complex Structures, and Singular Limits*, Erlangen, Germany

Nov. 2026 Section: Partial Differential Equations in *The 2026 Academic Annual Meeting of Chinese Mathematical Society*, Xi'an, China

Sept. 2026 Minisymposia: *Phase Transitions and Pattern Formation* and *Phase Field Methods in Real-World Applications* at the 17th International Conference on *Free Boundary Problems*, Berlin, Germany

July 2026 Special Sessions *Phase-field models and their singular limits* and *Mathematical Insights into Phase-Field Models* at *The 15th AIMS conference*, Athens, Greece

May 2026 Workshop on *PDEs and Stochastic PDEs in Material Science and Statistical Mechanics*, CUHKSZ, Shenzhen, China

Jan. 2026 International Conference on *Variational Methods in Fluid and Solid Mechanics*, Tsinghua Sanya International Mathematics Forum, Sanya, China

Nov. 2025 Workshop on *Nonlinear Partial Differential Equations*, Soochow University, Suzhou, China

June. 2025 *Mathematics with Applications 2025*, Funchal, Madeira, Portugal

Mar. 2025 *Mathematics Integrated Seminar*, School of Mathematical Sciences, Fudan University, China

Oct. 2024 *Youth Seminar on Differential Equations*, Anhui Normal University, Wuhu, China

July 2024 Workshop on *Partial Differential Equations and Its application*, Southwestern University of Finance and Economics, Chengdu, China

Apr. 2024 *Youth Seminar on Differential Equations*, Anhui Normal University, Wuhu, China

Nov. 2023 Seminar on *Partial Differential Equations*, Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic

Oct. 2023 Seminar of the Work Group *Nonlinear Partial Differential Equations*, Karlsruhe Institute of Technology, Karlsruhe

Aug. 2023 *ICIAM 2023*, 10th International Congress on Industrial and Applied Mathematics, Waseda University, Tokyo, Japan

Minisymposium *Modeling and Computation for Interface Dynamics in Fluids and Solids*

Aug. 2023 *Mathematics Integrated Seminar*, School of Mathematical Sciences, Fudan University, China

Dec. 2022 SEU Yau Center *Geometry and PDE* Weekend Seminar, Nanjing, China (online)

Apr. 2022 Working Seminar on Partial Differential Equations, Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic

○ Participants and talks

Nov. 2024 Conference on *Mathematics of fluids in motion: Recent results and trends*, CIRM, Marseille Luminy, France

Small talk

Oct. 2024 *The Seventh National Doctoral Forum on Partial Differential Equations*, Shenzhen University, Shenzhen, China

Contributed talk

June 2024 *Equadiff 2024*, Karlstad, Sweden

Contributed talk

May 2024 *The 5th International Symposium on Biological Mathematics and Medical Applications*, Nanjing, China

Contributed talk

May 2024 *11th Youth Forum of Partial Differential Equations*, Wuhu, China

Nov. 2023 International Workshop *Vector- and Tensor-Valued Surface PDEs*, Dresden

Contributed talk

Oct. 2023 Oberwolfach Seminar: *Recent Topics on the Navier-Stokes Equations*, Oberwolfach

Small talk

Sept. 2023 11th GAMM Workshop on *Analysis of PDEs*, Eichstätt

Contributed talk

Sept. 2023 Workshop on *Energetic Methods for Multi-Component Reactive Mixtures*: Modelling, Stability, and Asymptotic Analysis, WIAS Berlin

Contributed talk

June 2023 Shocking Developments: *New Directions in Compressible and Incompressible Flows*: A Conference in Honor of Alexis Vasseur's 50th Birthday, Max Planck Institute for Mathematics in the Sciences, Leipzig

May 2023 The *GAMM* Annual Meeting 2023, Dresden

Contributed talk in Session *Applied Analysis*

Mar. 2023 Workshop on *Analysis of PDEs*, Karlsruhe

Aug. 2022 Conference on *Mathematical Fluid Mechanics*, Prague, Czech Republic

Contributed talk

Aug. 2022 The *GAMM* Annual Meeting 2022, Aachen

Contributed talk in Session *Applied Analysis*

Aug. 2022 Hausdorff School on *Geometric Analysis and Nonlinear PDEs*, Bonn

Poster

June 2022 Lake Como School of Advanced Studies *Mathematical Models for Bio-Medical Sciences*, Como, Italy

Contributed talk

May 2022 International Symposium on *Biological Mathematics and Medical Applications*, NUIST, Nanjing, China (online)

Contributed talk at Graduate Forum

May 2022 Conference *Nonlinear PDEs in Fluid Dynamics*, CIRM, Marseille Luminy, France

Apr. 2022 Research school *Mathematical Advances in Geophysical Flows*, CIRM, Marseille Luminy, France

Small talk

Mar. 2022 SIAM Conference on *Analysis of Partial Differential Equations*, Berlin (online)

Mar. 2022 Workshop on *Mathematical Methods for Complex Phenomena*, Regensburg

Contributed talk

Mar. 2022 Workshop on *PDEs for Surfaces and Interfaces*, Regensburg

Oct. 2021 9th GAMM Workshop on *Analysis of PDEs*, Freiburg

Oct. 2021 Annual Meeting of RTG 2339 *IntComSin*, Weltenburg (talk)

Sept. 2021 International Conference *Free Boundary Problems*, Berlin (online)

Aug. 2021 Summer School *Fluids under Control*, Prague (online)

Aug. 2021 Germany-Japan Workshop on *Problems Related to Free Boundaries and Moving Contact Lines*, Tokyo (online)

Oct. 2020 TES Student Compact Course *Variational Methods for Fluids and Solids*, Berlin (online)

Teaching

- **Instructor**

Fall 26/27 Modern Analysis (for graduate students)
Fall 26/27 Calculus I-1 (for NNU-Macquarie joint institute)
Spring 26 Calculus I-2 (for NNU-Macquarie joint institute)
Fall 25/26 Ordinary Differential Equations (for math students)
Calculus I-1 (for NNU-Macquarie joint institute)
Spring 25 Advanced Mathematics II-2 (for engineering students, only two weeks)

- **Teaching Assistant**

Fall 24/25 Mathematical Analysis III (for math students)
Spring 24 Mathematical Analysis II (for math students)

Journal Referee

Acta applicandae mathematicae, Applied mathematical modelling, Chaos, solitons and fractals, Communications in Partial Differential Equations, Journal of elliptic and parabolic equations, Mathematical methods in the applied sciences, Physics of fluids.

Honors & Awards

Nov. 2021 **Excellent academic degree master's thesis** of Jiangsu Province, Jiangsu Academic Degree Committee
Oct. 2019 **National Scholarships for Graduate Students**, Ministry of Education of the People's Republic of China
Oct. 2019 **First Prize**, Postgraduate Scholarship, Nanjing University of Information Science and Technology
Oct. 2019 **National Second Prize**, 16th National Graduate Mathematical Modeling Contest
June 2018 **Outstanding Graduates**, Nanjing University of Information Science and Technology
Apr. 2017 **Meritorious Winner**, Mathematical Contest In Modeling/Interdisciplinary Contest In Modeling
Oct. 2016 **National First Prize**, Contemporary Undergraduate Mathematical Contest in Modeling
Nov. 2015 **First Prize**, Merit Scholarship, Nanjing University of Information Science and Technology

Languages & Skills

Languages Chinese Mandarin (Native)
Yancheng dialect (Native)
English (Fluent)
German (Elementary)
Programing MATLAB, Mathematica, SPSS

References & External Links

References

- Prof. Dr. **Helmut Abels**
  Fakultät für Mathematik
  Universität Regensburg
  Universitätsstraße 31
  93053 Regensburg, Germany
  helmut.abels@ur.de

- Prof. Dr. **Harald Garcke**
  Fakultät für Mathematik
  Universität Regensburg
  Universitätsstraße 31
  93053 Regensburg, Germany
  harald.garcke@ur.de

- Prof. **Wenjun Liu**
  School of Mathematics and Statistic
  Nanjing University of Information Science and Technology
  Ningliu Road 219
  Nanjing 210044, China
  wjliu@nuist.edu.cn

- RNDr. **Šárka Nečasová**, DSc.
  Institute of Mathematics
  Czech Academy of Science
  Žitná 25
  115 67 Praha 1, Czech Republic
  matus@math.cas.cz

- PD Dr. **Maria Neuss-Radu**
  Department Mathematik
  FAU Erlangen-Nürnberg
  Cauerstraße 11
  91058 Erlangen, Germany
  maria.neuss-radu@math.fau.de

External Links

- School of Mathematical Sciences, Nanjing Normal Unversity
- Research Training Group 2339 (RTG 2339) *Interfaces, Complex Structures, and Singular Limits*
- GAMM Activity Group *Analysis of Partial Differential Equations*
- Fakultät für Mathematik, Universität Regensburg